ECE343 Spring 2025

Objective 8.1: Displacement Current



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

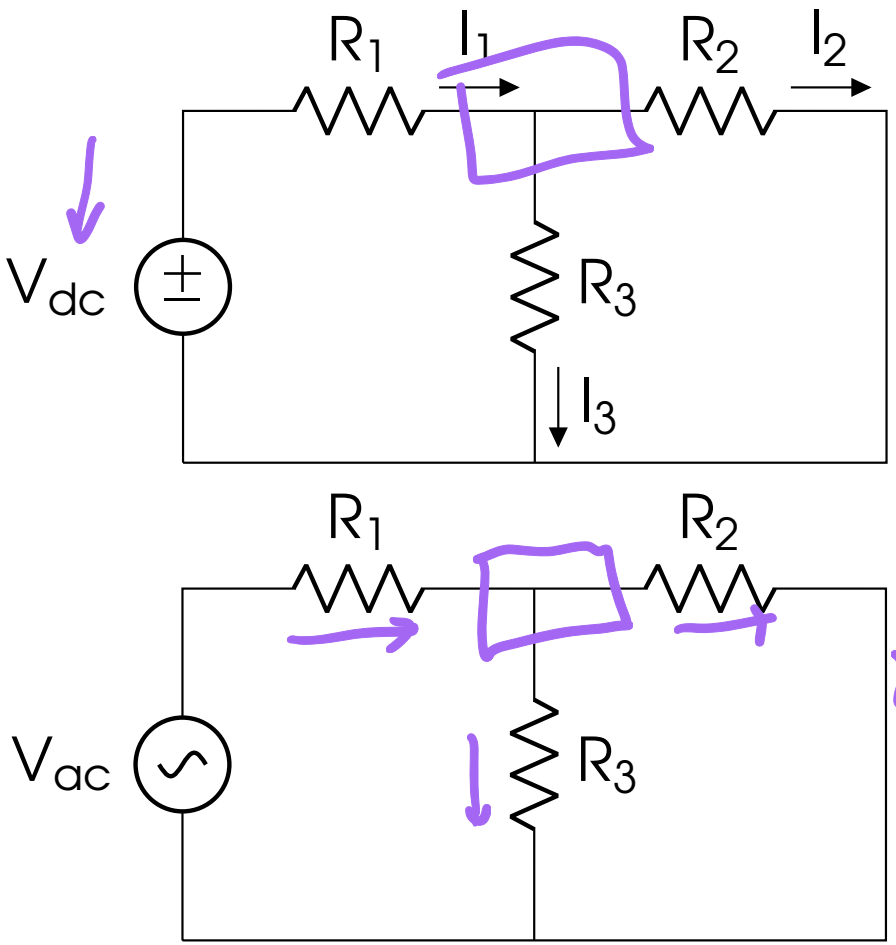
Ch 8 Learning Objectives

7.1 I can articulate the significance of the displacement current and calculate it, given a time-varying electric field.

7.2 I can articulate the meaning of and calculate the relaxation time for a variety of materials.

7.3 I know the forms of Maxwell's equations for time-varying fields and use them to calculate electric and magnetic fields for a variety of physical systems.

Displacement Current



KCL: $I_1 - I_2 - I_3 = 0$

$I = -\frac{dQ}{dt}$

$\text{div.} \int \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \iiint \rho_v \, dV$

$\iiint \nabla \cdot \vec{J} \, dV = \iiint \left(-\frac{d}{dt}\right) \rho_v \, dV$

$$\nabla \cdot \vec{J} = -\frac{d\rho_v}{dt}$$

current continuity

Insight #1: Relaxation Time

$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$

$\vec{J} = \sigma \vec{E}$ → ohm's law

$\nabla \cdot \sigma \vec{E} = -\frac{\partial \rho_v}{\partial t}$

$\nabla \cdot \vec{E} = -\frac{1}{\sigma}\frac{\partial \rho_v}{\partial t}$

Gauss: $\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon}$

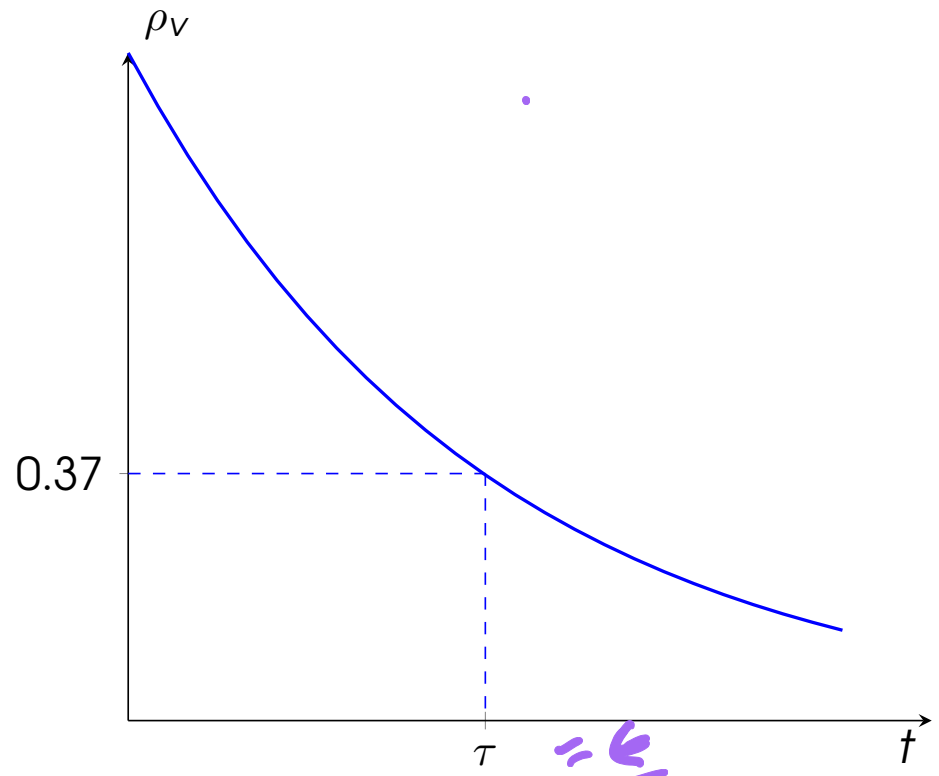
$\frac{1}{\sigma}\frac{\partial \rho_v}{\partial t} + \frac{\rho_v}{\epsilon} = 0$

$\Rightarrow \rho_v = \rho_0 e^{-t/\tau}$

$-\frac{\rho_0}{\tau} e^{-t/\tau} + \frac{\sigma \rho_0}{\epsilon} e^{-t/\tau} = 0$

$-\frac{1}{\tau} + \frac{\sigma}{\epsilon} = 0 \qquad \tau = \frac{\epsilon}{\sigma}$

$\rho_v = \rho_0 e^{-t/\tau}$



$e^{-1} =$

$\tau = \frac{\epsilon}{\sigma}$

Relaxation Time: Examples





Polystyrene

$\sigma = 10^{-17}$S/m and $\varepsilon = 2.56\varepsilon_0$

$\tau = \frac{(2.56)(8.854\times10^{-12}\,\text{F/m})}{10^{-17}\,\frac{1}{\Omega\text{m}}}$

$= 2.27\times10^{6}\,\text{s} \approx 26\,\text{day}$

Copper

$\sigma = 5.8 \times 10^{7}$S/m and $\varepsilon = \varepsilon_0$

$\tau = 1.5\times10^{-19}\,\text{s}$

Insight #2: Ampere's Law

$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$

Ampere: $\nabla \times \vec{H} = \vec{J}$

$\nabla \cdot (\nabla \times \vec{E}) = \nabla \cdot \vec{J}$

$\stackrel{?}{=}$ ↓ c.c.

$0 = -\frac{\partial \rho_v}{\partial t}$

Gauss: $\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon}$

$\nabla \cdot \vec{J}_d = \frac{\partial}{\partial t} \nabla \cdot \epsilon \vec{E}$

$\nabla \cdot \vec{J}_d = \nabla \cdot \left(\frac{\partial}{\partial t} \epsilon \vec{E} \right)$

$\vec{J}_d = \frac{\partial}{\partial t} \epsilon \vec{E} = \frac{\partial \vec{D}}{\partial t}$

Ohm's → $\vec{J}_c = \sigma \vec{E}$

James C. Maxwell

$\vec{J} = \vec{J}_c + \vec{J}_d$

condution → $\vec{J}_d$ ← displacement

$\nabla \cdot \vec{J} = \nabla \cdot \vec{J}_c + \nabla \cdot \vec{J}_d = 0$

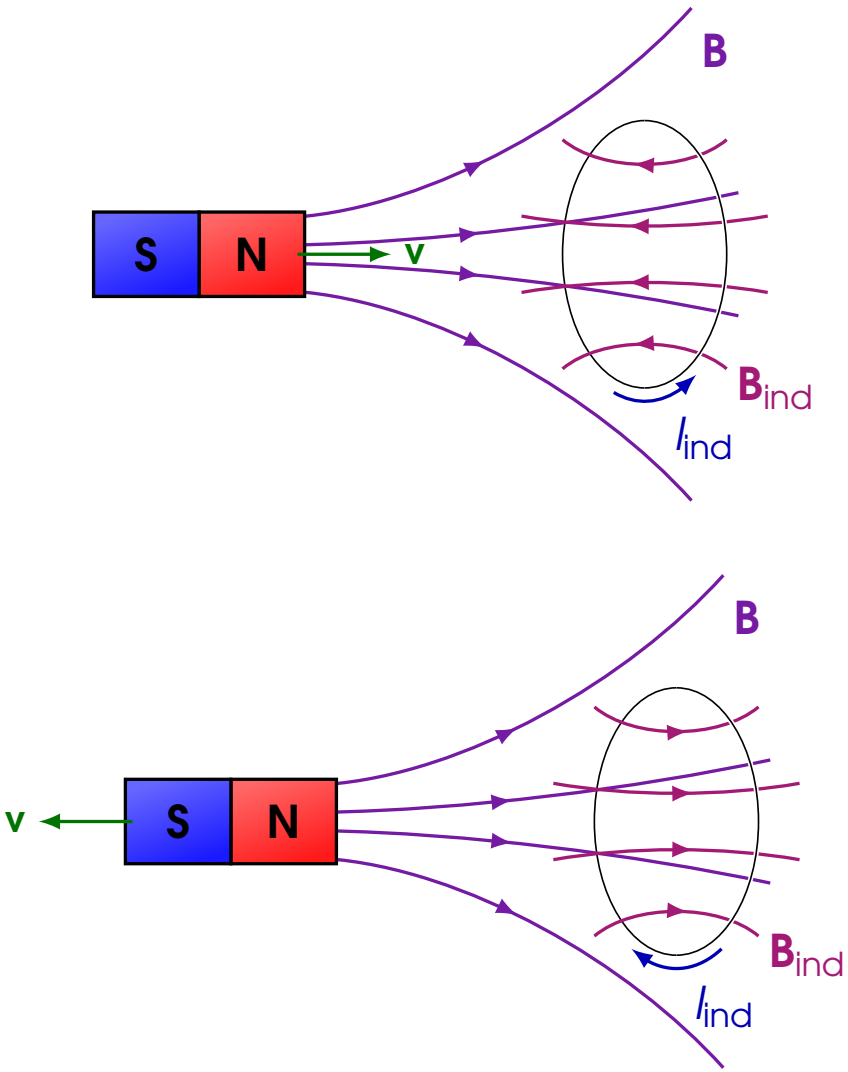
$\nabla \cdot \vec{J}_c = -\nabla \cdot \vec{J}_d = -\frac{\partial \rho_v}{\partial t}$

$\nabla \cdot \vec{J}_d = \frac{\partial \rho_v}{\partial t}$

Ampere's Law

$$\nabla \times \vec{H} = \vec{J}_c + \vec{J}_d = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

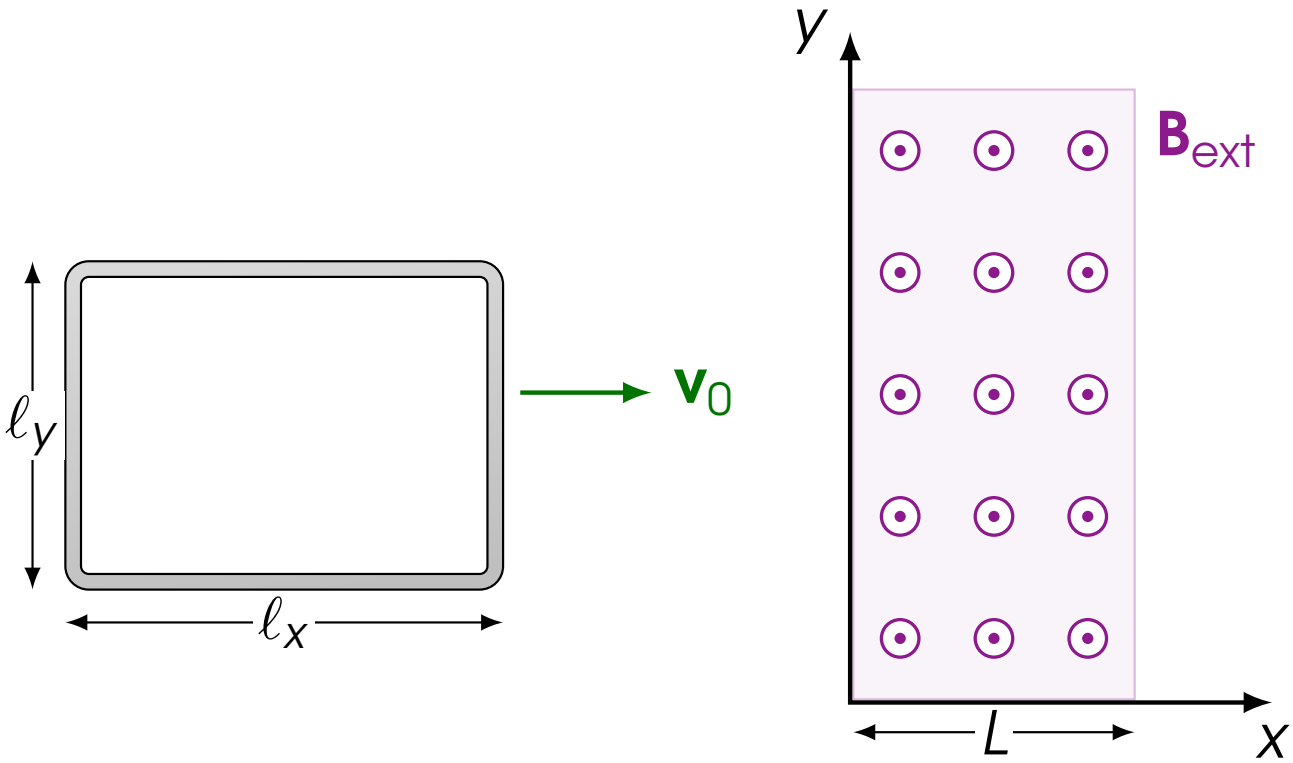
Lenz's Law



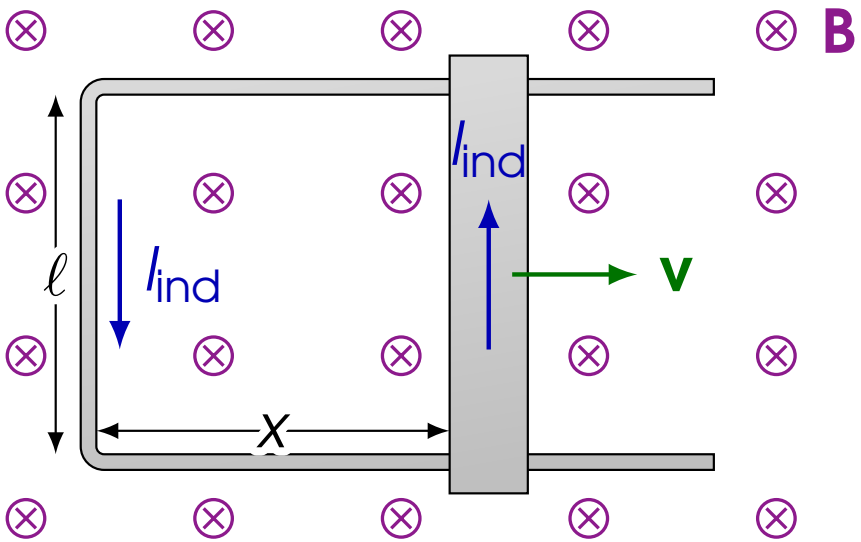
Lenz's Law

A current is induced by a change in the _____ magnetic field. The induced _____ results from an induced magnetic field opposed to the impressed _____ field.

Lenz's Law



Lenz's Law



Faraday's Law

$\Psi_m = \iint \vec{B} \cdot d\vec{S}$ $\Psi_E = \iint \vec{D} \cdot d\vec{S}$

Faraday's Law is the mathematical form of Lenz's Law:

Ψ_m is total Magnetic Flux

$$V = -\frac{\partial \Psi_m}{\partial t} = -\frac{\partial}{\partial t}\iint \vec{B} \cdot d\vec{S}$$

Ampere: $\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \epsilon \vec{E}}{\partial t}$ ($\vec{J}_c$, $\vec{J}_d$)

Electric potential comes from the electric field:

$V = \int \vec{E} \cdot d\vec{L} = -\frac{\partial}{\partial t}\iint \vec{B} \cdot d\vec{S}$

$\iint (\nabla \times \vec{E}) \cdot d\vec{S} = \iint \left(-\frac{\partial \vec{B}}{\partial t}\right) \cdot d\vec{S}$

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Faraday's Law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$